Arduino Code:

sketch_oct15c | Arduino IDE 2.3.6

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Arduino UNO

sketch_oct15c.ino

```
#include <Wire.h>

#define LM75_ADDRESS 0x48
#define PULSE_PIN 8

unsigned long pulseCount = 0;
unsigned long lastTempTime = 0;
unsigned long lastBpmTime = 0;

int previousPulseState = LOW;

float readTemperature()
{
  Wire.beginTransmission(LM75_ADDRESS);
  Wire.write(0x00);
  Wire.endTransmission();

  Wire.requestFrom(LM75_ADDRESS, 2);

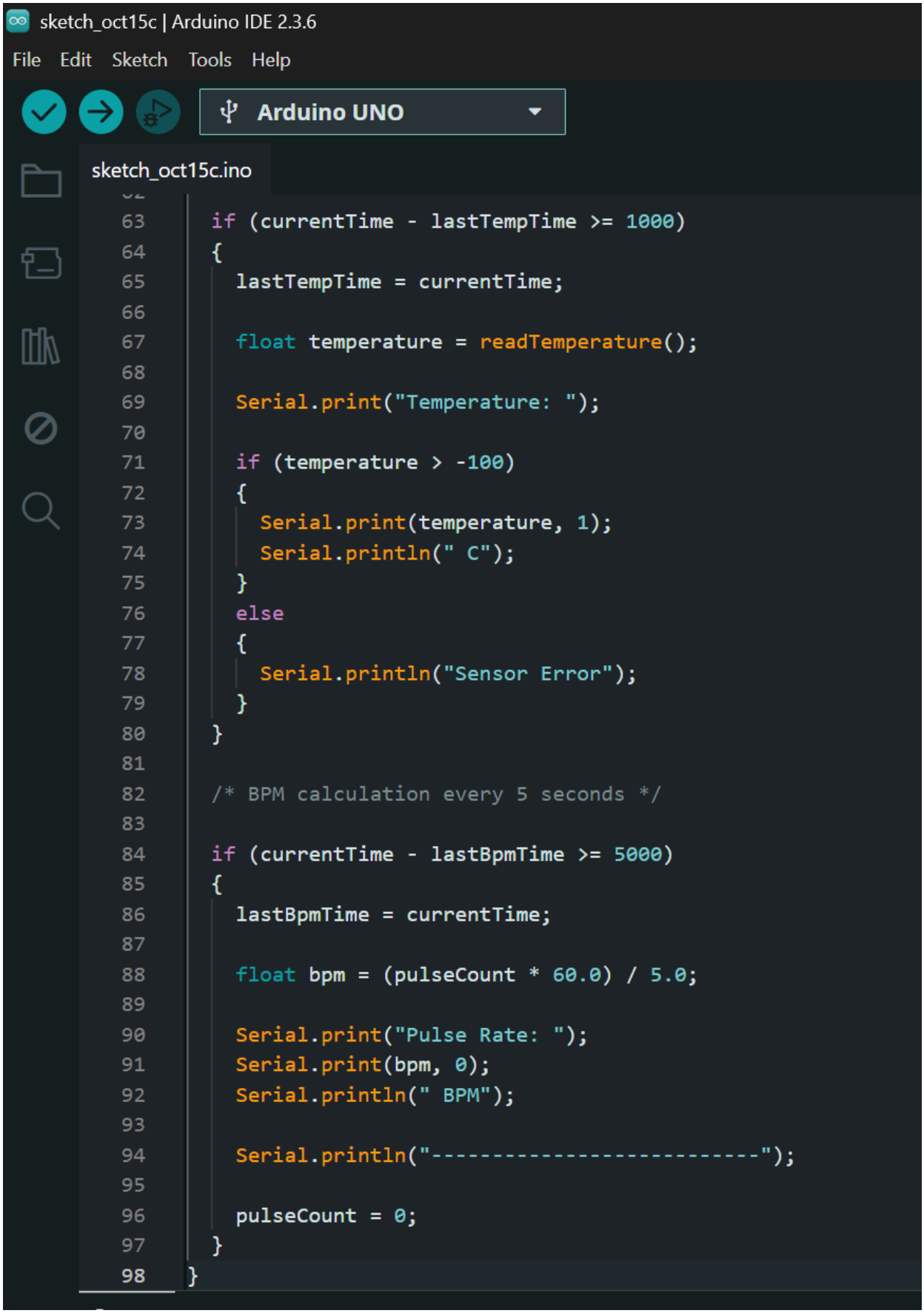
  if (Wire.available() >= 2)
  {
    byte msb = Wire.read();
    byte lsb = Wire.read();

    int16_t rawTemperature =
      ((int16_t)msb << 8) | lsb;

    return rawTemperature / 256.0;
  }

  return -999.0;
}

void setup()
```




```
  if (currentTime - lastTempTime >= 1000)
  {
    lastTempTime = currentTime;

    float temperature = readTemperature();

    Serial.print("Temperature: ");

    if (temperature > -100)
    {
      Serial.print(temperature, 1);
      Serial.println(" C");
    }
    else
    {
      Serial.println("Sensor Error");
    }
  }

  /* BPM calculation every 5 seconds */

  if (currentTime - lastBpmTime >= 5000)
  {
    lastBpmTime = currentTime;

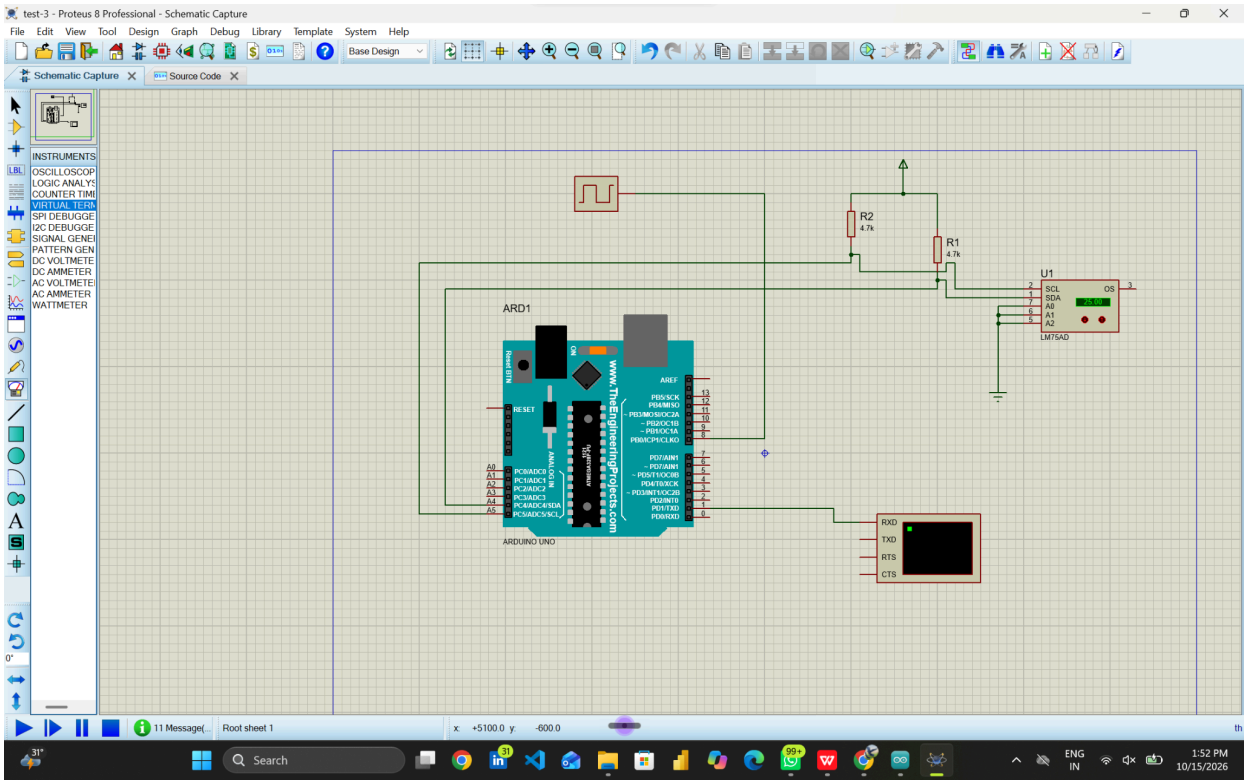
    float bpm = (pulseCount * 60.0) / 5.0;

    Serial.print("Pulse Rate: ");
    Serial.print(bpm, 0);
    Serial.println(" BPM");

    Serial.println("--------------------------");

    pulseCount = 0;
  }
}
```

Diagram:



Output:

